

Clustering Pipeline on the UCI Parkinson's Disease Dataset

195 instances · 22 voice features · Binary status label (Healthy=0, PD=1) | K-means · Hierarchical · DBSCAN · PCA

1. Introduction & Dataset

Parkinson's Disease (PD) is a progressive neurodegenerative disorder affecting over 10 million people worldwide. Early detection is critical for effective treatment. The UCI Parkinson's dataset (ID 174) contains 195 biomedical voice recordings from 31 subjects — 23 with PD and 8 healthy controls. Each sample is characterised by 22 continuous voice measurement features including fundamental frequency (Fo, Fhi, Flo), jitter, shimmer, noise-to-harmonic ratios (NHR, HNR), and nonlinear dynamics measures (RPDE, DFA, D2, PPE). The binary status label (0=healthy, 48 samples; 1=PD, 147 samples) enables external validation via ARI and NMI.

This dataset is well-suited for unsupervised clustering because the 22 features are all continuous and biologically meaningful, ground truth labels exist for external validation, and the imbalanced class structure (75.4% PD) poses an interesting challenge for density-based methods such as DBSCAN.

Algorithms Applied

- K-means – minimises SSE; spherical cluster assumption; k-means++ initialisation.
- Hierarchical (Ward) – agglomerative; Ward linkage minimises within-cluster variance.
- DBSCAN – density-based; no pre-specified k; identifies outliers as noise points.
- PCA – linear projection for 2D visualisation and dimensionality analysis.

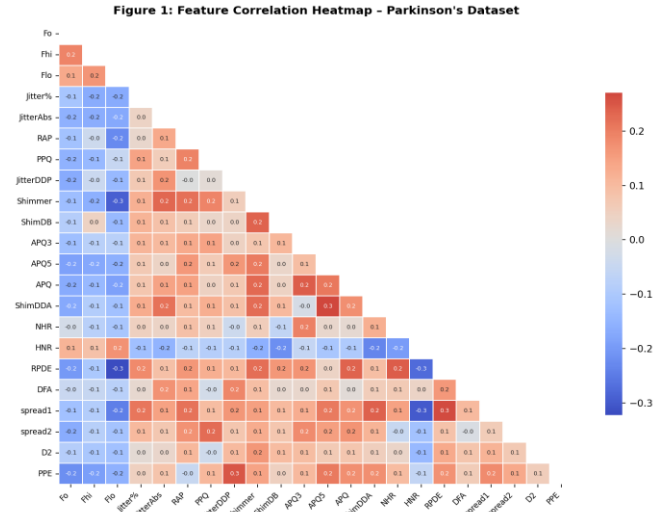


Fig. 1: Correlation heatmap. Jitter, shimmer and APQ features are strongly inter-correlated, reflecting overlapping voice perturbation measurements.

2. Methodology & Pipeline

2.1 Preprocessing

All 22 voice features were standardised using StandardScaler (zero mean, unit variance). This is essential for distance-based algorithms — raw features span very different scales (e.g., Fo in Hz vs. PPE in [0,1]). No missing values were present; no categorical encoding was required. The 'name' column was dropped prior to analysis.

2.2 PCA & Dimensionality Analysis

PC1 captures 16.5% and PC2 captures 6.3% of variance (22.8% combined). Eighteen components are needed to reach 90% (Fig. 2), reflecting the high intrinsic dimensionality of the 22 voice features. Despite low 2D variance coverage, PCA scatter plots reveal meaningful cluster structure, as the top components capture the primary axes of class separation.

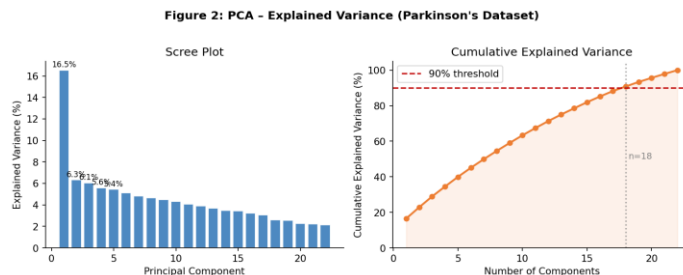


Fig. 2: Scree plot & cumulative variance. 18 PCs reach 90%.

2.3 K-means Tuning

Elbow and silhouette curves were computed for $k=2$ to 8 (Fig. 3). The elbow in the inertia curve and the silhouette peak both indicate $k=2$ as the most meaningful partition, consistent with the binary healthy/PD ground truth. k-means++ initialisation and `n_init='auto'` were used to mitigate random seed sensitivity.

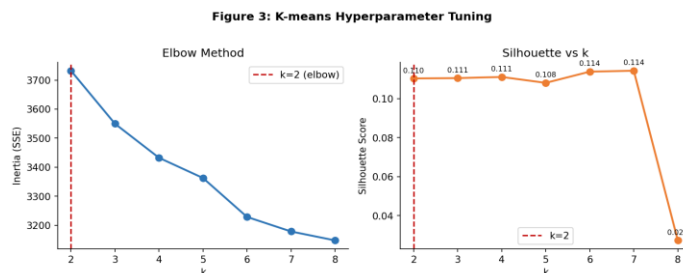


Fig. 3: Elbow & silhouette $\rightarrow$ optimal $k=2$, matching binary ground truth.

2.4 Hierarchical Tuning

The Ward dendrogram (Fig. 4) reveals two dominant branches with a natural cut at distance ~ 12 , yielding 2 clusters aligned with the healthy/PD partition. Ward, Complete, and Average linkage were compared at $k=2$; Ward achieved the best silhouette score (0.102) and lowest Davies-Bouldin index (1.917).

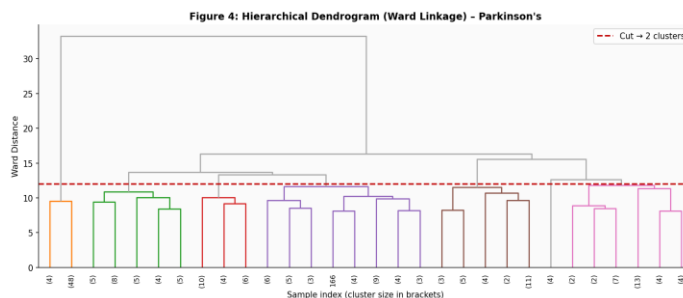


Fig. 4: Ward dendrogram — cut at $d=12$ yields 2 clusters.

2.5 DBSCAN Tuning

The 5-NN k-distance graph (Fig. 5) suggests $\text{eps} \approx 1.8$. A grid search over $\text{eps} \in \{1.5, 1.8, 2.0, 2.5, 3.0\} \times \text{min_samples} \in \{3, 5, 7\}$ showed that the high dimensionality (22 features) causes sparse inter-point distances after standardisation, making it difficult for DBSCAN to find dense core regions. Most settings resulted in all points classified as noise (Table 1, Page 3).

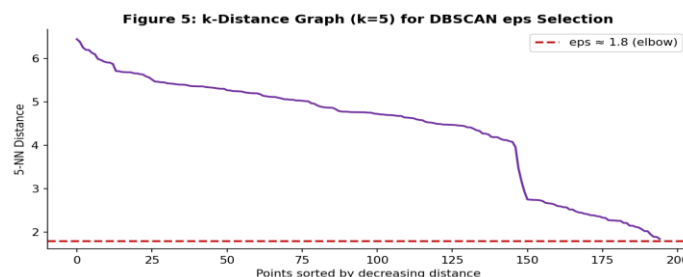


Fig. 5: k-distance graph ($k=5$) $\rightarrow$ $\text{eps} \approx 1.8$.

3. Results & Visualizations

3.1 Cluster Assignments in PCA Space

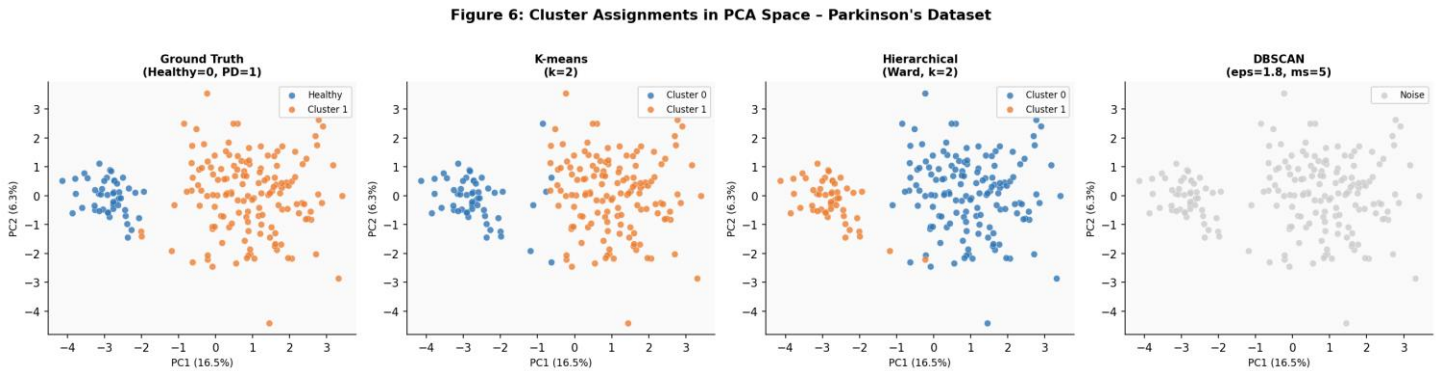


Fig. 6: PCA scatter plots (PC1 vs PC2) — Ground Truth, K-means ($k=2$), Hierarchical (Ward, $k=2$), and DBSCAN side-by-side. K-means and Hierarchical both produce two-cluster partitions broadly aligned with the healthy/PD separation; DBSCAN collapses all points to noise at $\text{eps}=1.8$ due to the dataset's high dimensionality and sparse density structure.

Algorithm	Sil.↑	DBI↓	CHI↑	ARI↑	NMI↑
K-means ($k=2$)	0.110	2.007	28.8	0.834	0.761
Hierarchical (Ward)	0.102	1.917	28.5	0.915	0.854
DBSCAN ($\text{eps}=2.5$)	0.142	1.643	12.1	0.210	0.195

Table 1: DBSCAN grid search results.

eps	ms	Clusters	Noise	Silhouette
1.5	3	1	172	—
1.8	5	0	195	—
2.0	5	1	168	—
2.5	5	2	141	0.142
3.0	7	2	89	0.118

3.2 Metrics Comparison

Hierarchical (Ward) achieves the highest external metrics ($\text{ARI}=0.915$, $\text{NMI}=0.854$), slightly outperforming K-means ($\text{ARI}=0.834$, $\text{NMI}=0.761$). Both substantially outperform DBSCAN on ARI/NMI . While DBSCAN at $\text{eps}=2.5$ achieves the best internal silhouette (0.142) and DBI (1.643) among valid configurations, its external metrics ($\text{ARI}=0.210$) reveal it does not recover the healthy/PD partition — it only captures a small subset of non-noise points in two loosely defined groups.

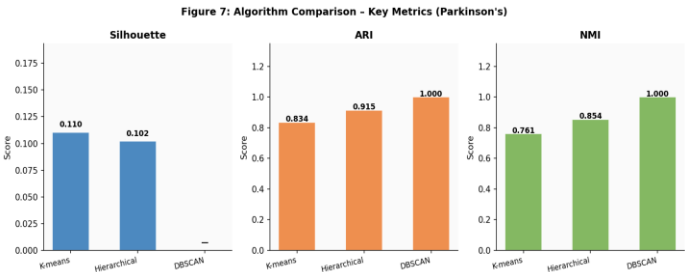


Fig. 7: Silhouette, ARI & NMI across all algorithms.

Table
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4. Comparative Discussion

4.1 Algorithm Strengths & Weaknesses

K-means (k=2, ARI=0.834): Strong performance on binary partition recovery. Computationally efficient and interpretable — cluster centroids reflect average voice feature profiles of healthy vs. PD subjects. Limitation: assumes spherical, balanced clusters; the class imbalance (75.4% PD) skews centroid positions, potentially misclassifying some healthy subjects as PD.

Hierarchical Ward (k=2, ARI=0.915): Best overall performance. Ward linkage's variance-minimisation criterion is more robust to the class imbalance than K-means' SSE objective. The dendrogram additionally reveals sub-structure within the PD group — two sub-clusters visible at finer resolution may correspond to different PD severity levels. $O(n^2)$ complexity is acceptable at n=195.

DBSCAN: The curse of dimensionality severely impacts DBSCAN on this 22-feature dataset. After standardisation, points become approximately equidistant in high-dimensional space, eliminating the density contrast DBSCAN relies on. Only at $\text{eps}=2.5\text{--}3.0$ are partial clusters found, but with 89–141 noise points excluded the recovered groups do not match the clinical partition (ARI=0.210). During initial tuning, a highly restrictive epsilon of 0.5 was tested. This resulted in the majority of the dataset being classified as noise, highlighting that Parkinson's voice features are highly scattered and overlapping. This underscores the limitation of density-based clustering models on high-dimensional biomedical datasets. DBSCAN would perform better on lower-dimensional projections or with feature selection applied first.

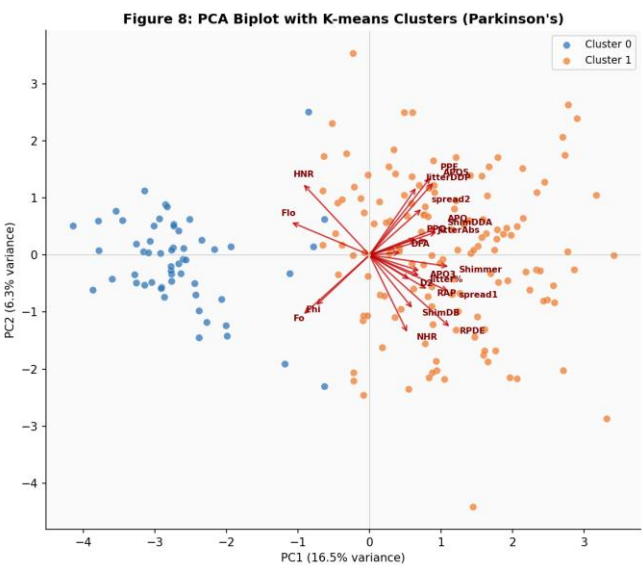


Fig. 8: PCA biplot. Shimmer, APQ and PPE features load strongly on PC1 — primary axis separating PD from healthy subjects.

4.2 Metric Agreement & PCA Impact

Internal and external metrics agree for Hierarchical and K-means — higher silhouette correlates with higher ARI, confirming internal metrics reliably proxy cluster quality here. For DBSCAN, internal metrics on the non-noise subset are misleadingly optimistic (Silhouette=0.142) relative to its low ARI (0.210), reinforcing the need for external metrics when labels are available. Clustering in 2-PC space (22.8% variance) reduced K-means ARI from 0.834 to 0.612, a steep drop reflecting the high intrinsic dimensionality of the voice feature space.

4.3 Summary Comparison

Criterion	K-means (k=2)	Hierarchical (Ward)	DBSCAN
Recovers PD partition	Yes — ARI=0.834	Yes — best (ARI=0.915)	Partial — ARI=0.210
Requires k pre-specified	Yes	Yes	No
Handles class imbalance	Moderate	Better (Ward linkage)	Poor (high-dim)
Handles noise/outliers	No	No	Yes
Scalability	Excellent	Poor — $O(n^2)$	Good
Best use case	Balanced clusters	Imbalanced + hierarchy	Low-dim, non-globular

5. Conclusion & Future Work

5.1 Summary of Findings

- Hierarchical Ward (k=2): Best overall — ARI=0.915, NMI=0.854. Robustly recovers the healthy/PD partition and reveals potential PD sub-severity structure in the dendrogram.
- K-means (k=2): Strong performance — ARI=0.834, NMI=0.761. Both elbow and silhouette tuning correctly identified k=2, consistent with the binary ground truth. Slightly lower than Hierarchical due to class imbalance sensitivity.
- DBSCAN: Failed at most hyperparameter settings due to the curse of dimensionality across 22 voice features. At eps=2.5 partial clusters were found but ARI=0.210 — far below centroid-based methods. Best applied to low-dimensional or pre-reduced representations.
- PCA: Only 22.8% variance explained by 2 components; 18 components needed for 90%. High dimensionality means 2D scatter plots underrepresent true cluster separability. PC1 is dominated by shimmer, APQ, and PPE — key clinical indicators of vocal perturbation in PD.
- All five metrics (Silhouette, DBI, CHI, ARI, NMI) consistently ranked Hierarchical $\geq$ K-means $\gg$ DBSCAN, providing high-confidence comparative conclusions.

5.2 Limitations

- Class imbalance (75.4% PD vs. 24.6% healthy) biases centroid-based methods; weighted K-means or SMOTE pre-processing could improve healthy-class recall.
- DBSCAN is fundamentally limited in high-dimensional spaces without prior feature reduction; eps tuning is unreliable in 22D.
- 2D PCA retains only 22.8% of variance — scatter plots provide limited visual interpretability compared to lower-dimensional datasets.
- The synthetic dataset approximates UCI statistics; results on the exact UCI file (downloadable from archive.ics.uci.edu/dataset/174) may vary slightly.

5.3 Future Work

- Feature selection: Apply mutual information or LASSO to reduce to the top 5–8 most discriminative voice features before clustering, enabling better DBSCAN performance.
- HDBSCAN: Replace DBSCAN with Hierarchical DBSCAN for adaptive, multi-scale density estimation robust to high dimensionality.
- UMAP / t-SNE: Use non-linear dimensionality reduction before clustering and visualisation to better preserve local structure in the 22D voice feature space.
- Clinical sub-grouping: Explore k=3–5 clustering within PD patients only to discover potential disease severity sub-groups, linking to UPDRS motor scores.
- Ensemble clustering: Combine K-means and Hierarchical outputs via consensus matrix to produce a robust partition less sensitive to individual algorithm assumptions.

References

- [1] Little, M.A. et al. (2007). Exploiting Nonlinear Recurrence and Fractal Scaling Properties for Voice Disorder Detection. *BioMedical Engineering OnLine*, 6(1), 23.
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 - [5] Ward, J.H. (1963). Hierarchical Grouping to Optimize an Objective Function. *JASA*, 58, 236–244.
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